

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
ML / Deep Learning	PyTorch, JAX, HuggingFace Transformers, NumPy, scikit-learn, autodiff, normalizing flows, diffusion, variational inference / natural-gradient optimization, mechanistic interpretability / activation steering
LLM / Eval	LLM agents, prompt design, evaluation methodology, anti-Goodharting, holdout protection, RAG, RRF, cross-encoder reranking, OpenAI / Groq APIs, Ollama
Systems & Data	Data pipelines, Apache Spark / PySpark, Spark UI, async services, distributed-data concepts, Linux profiling, FastAPI, REST APIs
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Research & Projects

Refusal Decay - LLM Behavior Steering & Interpretability

UMass CS 602 Research Project (graded full marks)

PyTorch, HuggingFace Transformers, Llama-3.1-8B-Instruct

- Ran a mechanistic-interpretability study of how prefilling attacks weaken refusal in **Llama-3.1-8B-Instruct**; extracted the residual-stream refusal direction via difference-in-means and **steered model behavior** with activation patching and additive direction injection through PyTorch forward hooks.
- Designed multi-seed random/orthogonal controls and a benign positive control, reporting a **clean null** (0/300 recovered) with bootstrap **95% CIs** and McNemar’s test, and validated heuristic labels against a secondary classifier (87.5% agreement).

AutoEval - Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, ML Evaluation, Git Automation

- Designed an evaluation-improvement agent that proposes queries and labels, scores them against multiple systems, and accepts only updates that improve **discriminative power** on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

Generative Models & Deep Learning

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented a **matrix-based reverse-mode autodiff engine from scratch** in NumPy, validated against JAX to machine precision; benchmarked **5 architectures across 3 optimizers** on CIFAR-10 and derived GD/SGD convergence guarantees.
- Implemented a **normalizing flow with coupling layers** and a **DDPM diffusion model** from scratch in JAX, deriving the ELBO and Gaussian-KL objectives via variational inference.

KG2RAG-Enhanced - Multi-Hop QA on HotpotQA

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, PyTorch | UMass CS 685 Advanced NLP (team)

- Led the **multi-view retrieval** component of a team KG-guided RAG pipeline (sub-question decomposition, per-view retrieval, RRF fusion + cross-encoder reranking + MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a full-stack **ML training-data pipeline / annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label noisy real-world data for production models; cut manual annotation by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.